electrical arrangement is in effect a capacitor, only a minimal amount of current is needed to charge the capacitor, giving us the first advantage of electrostatic speakers over electrodynamic speakers – the former is a high impedance device with a low current requirement, while the latter are low impedance devices, with higher current requirements. However, the main disadvantage, which has prevented their commercial production to quite an extent, is the requirement of a transformer which helps match impedance levels. Seeing as how the design of the transformer is almost always specific to a particular electrostatic speaker, it makes production of these varieties of speakers very tough and commercially unviable.

Then there are the piezoelectric speakers, which use the piezoelectric effect (the accumulation of electric charge in certain solid materials such as crystals, certain ceramics and lead alloys, in response to applied mechanical stress), coupled with the diaphragm design in other speaker systems to produce sound. The diaphragm is generally a piezoelectric material stuck on top of a regular diaphragm. When exposed to an electric field, this piezoelectric material will shrink or expand as charge is introduced or removed, and thus causes elastic deformation of the material perpendicular to the direction of the speaker. As soon as the electric field is removed from the piezoelectric material, it will return to its original shape. This flexing of the diaphragm is what produces a movement in the air molecules, eventually causing a chain of collisions that produces sound.

Magnetostatic speakers

In most cases, piezoelectric speakers can have extended high frequency output. However, because their frequency response is poor, and inferior to other available technologies, especially with regards to bass and midrange, they are used in applications where volume and high